

Practice Problems List-5 (Loops)

1. Write a program in C to print "Hello World" a hundred times.
2. Write a C program to print all positive integers from 1 to 100.
3. Write a C program to print all negative integers from -1 to -100.
4. Write a program to print "Hello World" ten times with serial numbers like:
 1. Hello World
 2. Hello World
 3. Hello World ...
5. Write a C program to print all positive integers from 5 to 500.
6. Write a C program to print all even integers from 1 to 100.
7. Write a C program to print all odd integers from 1 to 100.
8. Write a C program to print all integers divisible by 7 between 55 and 155
(inclusive).
9. Run a loop from 1 to 100, but print the numbers from 1 up to 25 only. (do not use
break or continue)
10. Run a loop from 5 to 105, but print the numbers from 5 up to 25 only. (use break)
11. Run a loop from 5 to 105, but print the numbers from 5 up to 25 only. (use
continue)
12. Run a loop from 1 to 100, but print the numbers from -1 to -100.
13. Write a C program to print all positive integers from 100 to 1.
14. Write a C program to print all even integers from 1 to 100 in reverse order. [100,
98, 96,, 2]
15. Write a C program to print the following series: 5 10 15 20 25 30 35 40 500

16. Write a C program to print the following series: 2, 4, 8, 16, (up to 100)
17. Write a C program to print the following series: 1000, 500, 250, 125,, 0
18. Take an integer number as input and print out its times table (multiplication table) as output.

Sample input: **5**

Sample output:

5x1=5

5x2=10

5x3 =15

.....

.....

5x10=50

19. Take input two integers **a** and **b** where both **a** and **b** are positive. Write a C program to print all natural numbers from a to b if $a \leq b$, or b to a if $b \leq a$.
20. Write a C program to print the sum of all natural numbers from 1 to 100.
21. Write a C program to calculate the sum of the following series: (take n as input)
- $$1^2 + 2^2 + 3^2 + 4^2 + 5^2 + \dots + n^2$$
22. Write a C program to print the sum of *cubes* of all natural numbers from 1 to n.
23. Write a C program to take a hundred integers as input using scanf() and output the number of even numbers among them.
24. Write a C program to take a hundred integers as input using scanf() and output the sum of them.
25. Write a C program to take a hundred integers as input using scanf() and output the largest of them.

26. Write a C program to take a hundred integers as input using `scanf()` and output the average of them in decimal up to two decimal points.
27. Write a C program to print all letters in the English alphabet.
28. Write a C program to take a hundred English letters as input using `scanf()` and output the number of vowels and consonants among them.
29. Write a C program to take a hundred characters as input using `scanf()` and output the number of uppercase letters, lowercase letters, digits, and special characters among them.
30. Write a C program to take five English letters as input using `scanf()` and toggle the letters while outputting them. [toggle = flipping the case]

Sample Input: Fahim

Sample Output: fAHIM